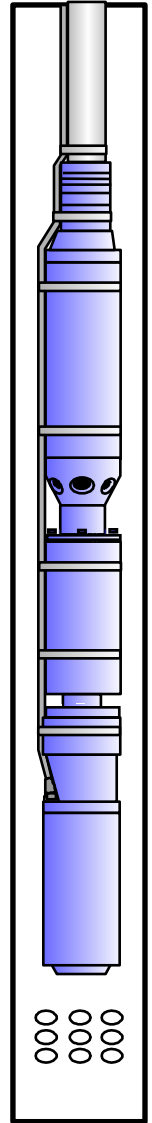




ESP Run Life Factors

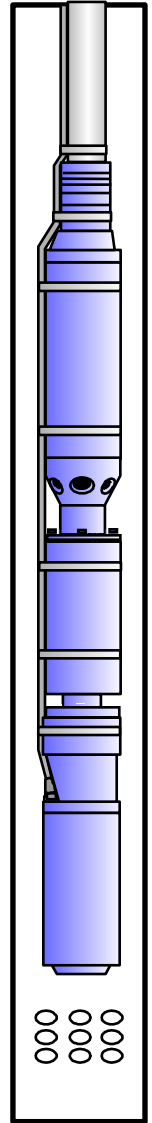
Run Life Factors

- The run life of an ESP is based on many factors
- Each well is different and may have a combination of factors that limit ESP run life
- The run life will be determined by the limiting factor in the well



Common Run Life Factors

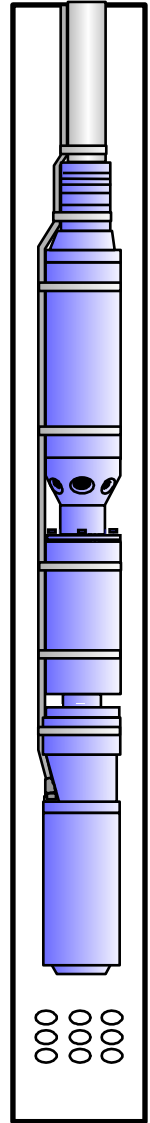
- Proper Sizing of Equipment
- Well (BHT) Temperature
- Free Gas
- Viscosity
- Corrosion
- Sand / Foreign Material Production
- Deposition Tendencies
- Electrical Failures
- Operational Problems
- Old Age



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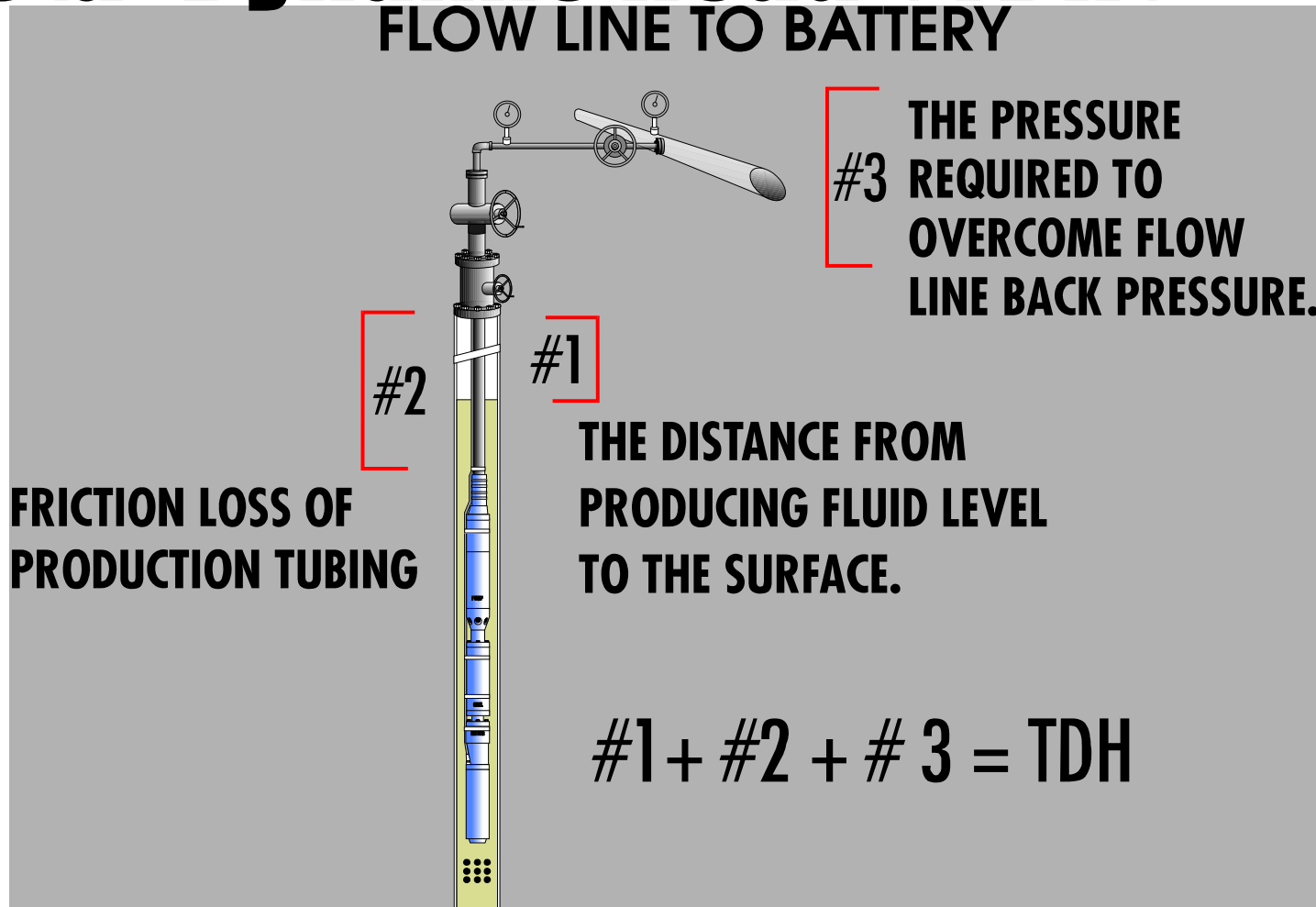
Proper Sizing

- Proper sizing of the ESP unit is the first factor in achieving a long run life
- The unit must be sized to operate within the recommended flow range
- Well productivity data must be accurate in order to properly size the equipment
- Improper sizing can cause the ESP to run outside of operating range causing accelerated pump wear
- Inaccurate fluid data can cause the BHP of the pump to be more than predicted which could cause motor overload, and eventually failure



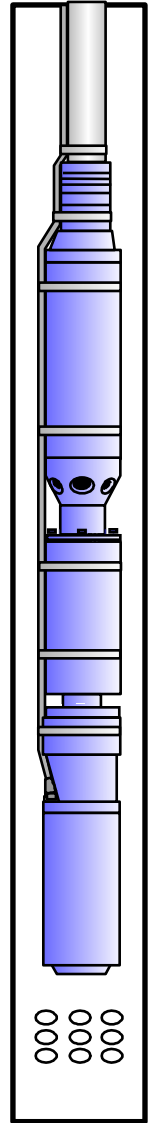
Total Dynamic Head (TDH)

FLOW LINE TO BATTERY



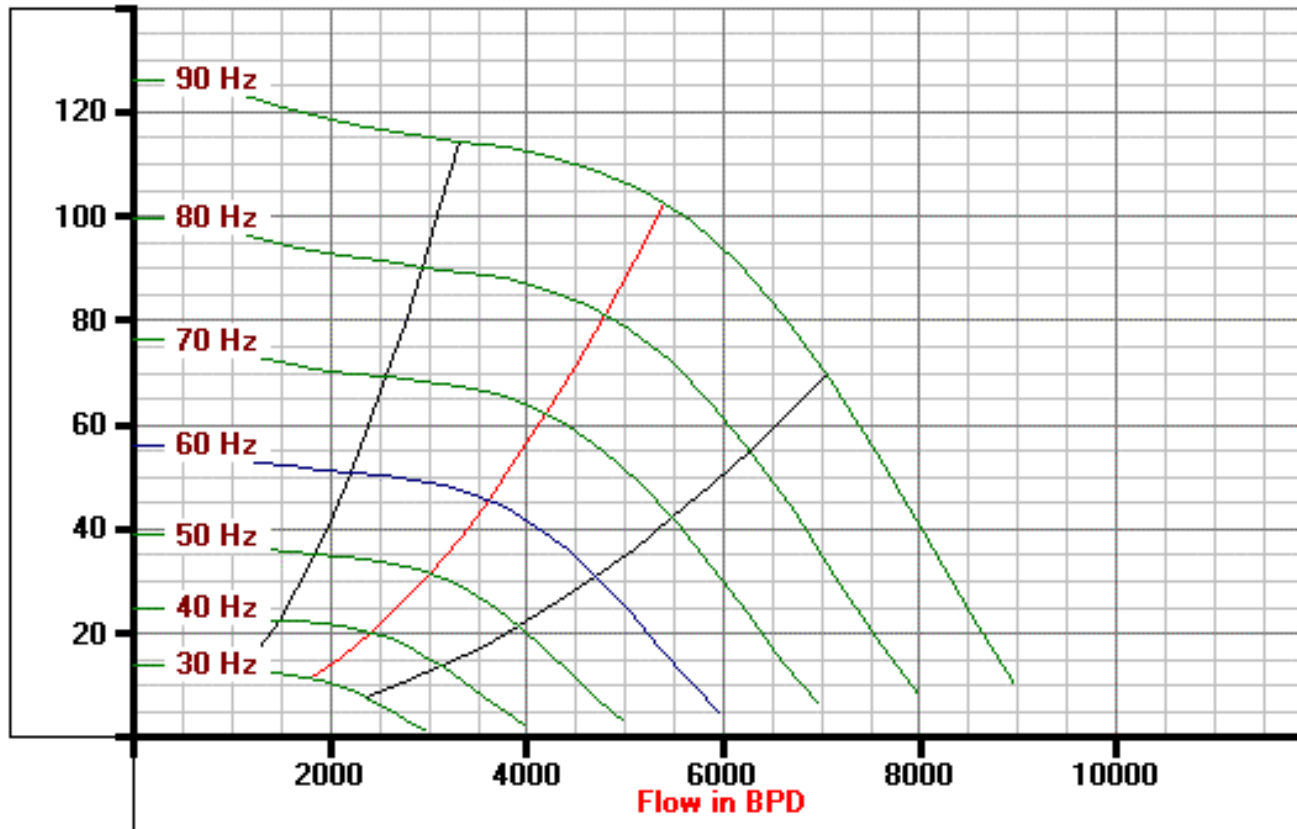
Proper Sizing Solutions

- Accurate reservoir and inflow performance data
- Accurate fluid properties information
- Computer models and correlations should reflect well parameters as closely as possible (average percent correlation error 5 - 15%)
- Use of a VSC can help to offset sizing inaccuracy by extending the operating range within limits



Variable Speed Pump Curve

Head in FT



Head vs. Flow, Variable Speed (30-90 Hz),
Single Stage, $S_g = 1.0$



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High Temperature

- Bottom hole temperatures greater than 220° F are considered high temperature applications for ESPs
- Motor operating temperature is also effected by
 - % load versus nameplate motor horsepower
 - Fluid velocity past the motor
 - % water, % oil, % gas of well fluid past motor
 - Power quality (unbalanced current, distorted wave form)
- The combination of all above factors determines the unit operating temperature



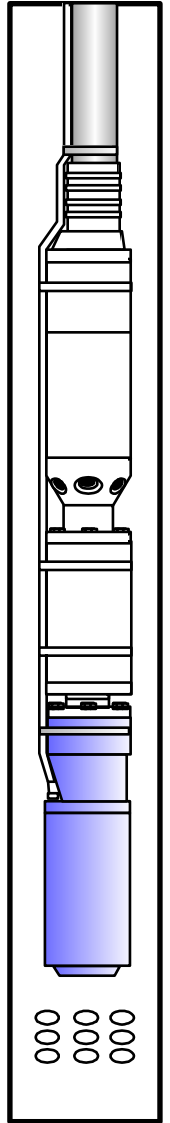
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High Temp Solutions

ESPs can run for long periods of time in high temperatures wells if the proper equipment is used. The following equipment features are recommended

High temp motor oil - retains viscosity at higher temperatures (also has good low temp properties)

- High temperature elastomers - EPDM cable insulation and jacket, O-rings, Aflas seal bags
- Special construction of rotor assembly in motor to insure proper bearing clearances
- De-rating motors for very high temperatures, if required



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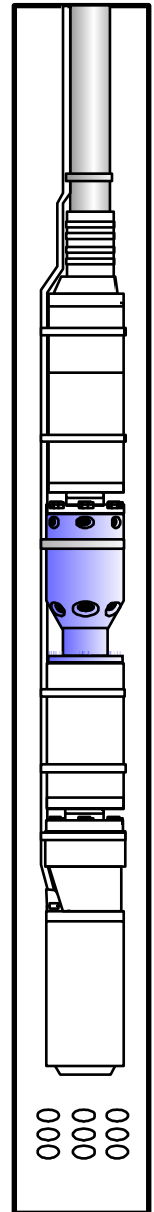
Free Gas

The presence of free gas can effect the ESPs in many ways...

- The pump flow will be reduced or completely stopped as the free gas increases. This is called “gas locking”
- The motor will run hotter as the fluid velocity decreases past the motor
- The fluid’s cooling properties will decrease as the free gas increases

Gassy Well Solutions ...

- Separate free gas before it enters the pump stages by rotary or reverse flow separators
- Utilize tapered pump stage designs to handle the increased gas volume through the pump



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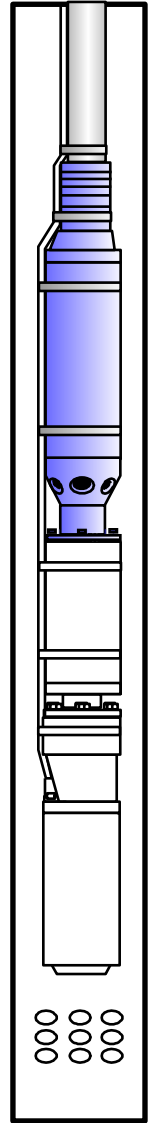
Viscosity

High fluid viscosity can cause many problems ...

- The specific gravity of the fluid increases, therefore increasing pump brake horsepower
- High viscosity also reduces the pumps ability to lift the fluid and its efficiency
- Viscous fluid produces more friction loss in the tubing causing the pump to work much harder

Viscosity Solutions ...

- Size pumps with higher flow stages and higher HP motors
- Dilute well fluid with low viscosity crude

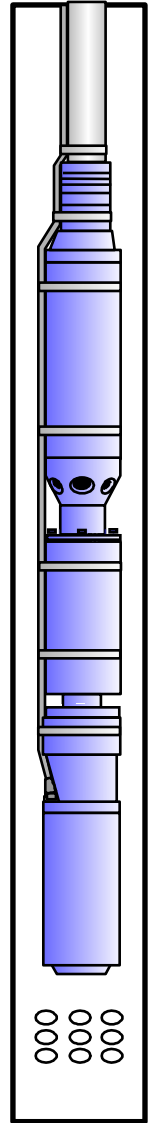


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Corrosion

Corrosive fluid effects ESPs when ...

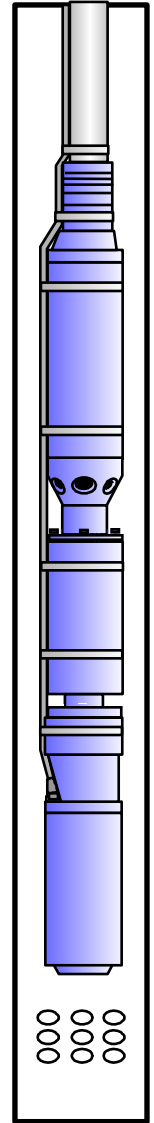
- CO_2 causes corrosion of housings, heads, bases, and fasteners of the downhole assembly
- CO_2 causes corrosion of galvanized cable armor on the power cable, connectors, and MLE
- H_2S chemically reacts with copper components causing cable conductors to disintegrate
- H_2S causes sulfide corrosion cracking with certain steels which effects shafts and bolts



Corrosion Solutions

For corrosive wells ESPs should have ...

- Corrosion resistant housings (9% Cr minimum)
- Stainless steel heads, bases, and fasteners
- Stainless Steel or Monel cable armor
- Monel or Inconel pump and seal shafts to address sulfide corrosion cracking
- Lead sheath cable for high H₂S environments (defined as above 3% & 180° F or greater)



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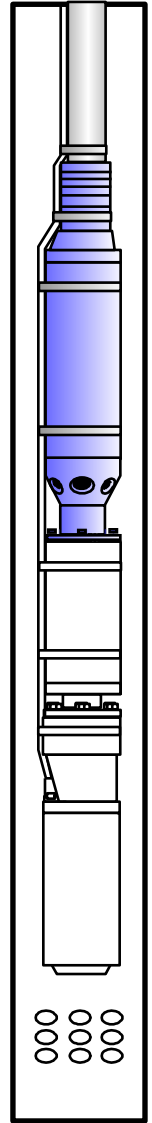
Sand Abrasion

The production of sand causes ...

- Abrasive wear on the pump stages
- Excessive shaft pump shaft vibration
- Mechanical seal leakage in the seal section
- Motor burns due to fluid migration

Solutions for sand production are ...

- Abrasion resistant pump design which provides for downthrust support and radial shaft stabilization
- Slow, steady increase in production of well on initial start up to limit inflow of unconsolidated sand



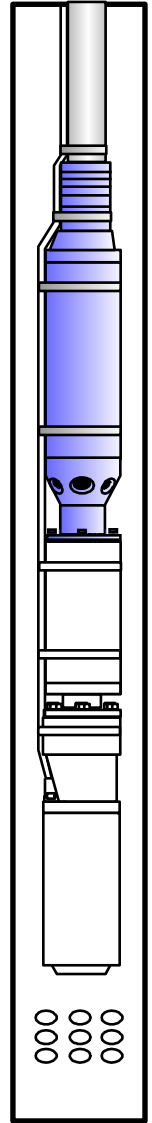
Foreign Material

The production of foreign material can cause ...

- Damage to pump stages if debris is harder than the pump stage material (unit fails similar to abrasion)
- Plugging of pump stage vane passages if debris is softer than pump stage material
- Low flow by the motor due to partially or totally plugged pump resulting in a burn of the motor or power cable

Foreign material solutions ...

- Thorough well clean out after each workover
- Slow, steady increase in production of well on initial start up to limit inflow of unconsolidated debris and foreign material
- Screens



Deposition

Deposition on pump stages cause high brake horsepower,
locked stages, &/or restrict pump or tubing

Types of deposition are ...

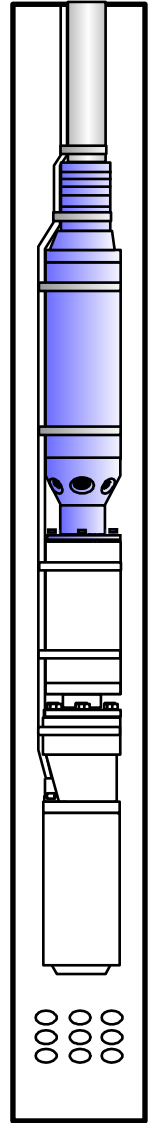
- Scale
- Asphaltenes
- Paraffin
- Hydrate / Ice Plugs

Deposition Solutions ...

- Chemical treatment
- Tubing heat (except for scale)
- Control pump intake pressure (except for hydrates)



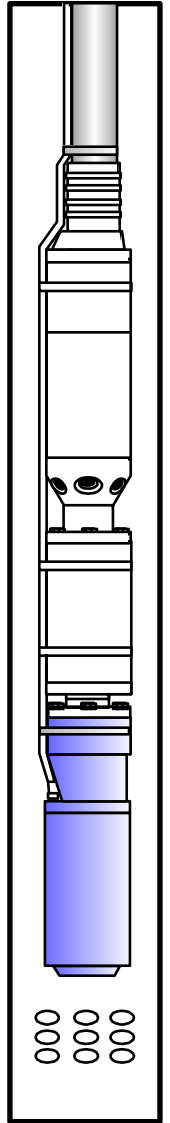
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Electrical Failures

Electrical failures are caused by factors such as ...

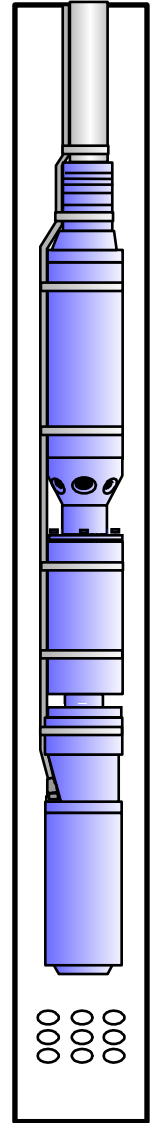
- Surface electrical or electronic component failure
- Poor Power such as voltage imbalance
- Cable failure due to decompression damage or voltage spikes
- Overload of the controller or transformer due to changes in downhole or unit conditions



Operating Practices

Poor operating practices can cause failure of ESPs. The most common are ...

- Operating the unit against the closed surface valve for an extended length of time (no flow by the motor will cause the motor or MLE to burn)
- Operating the unit in a no-flow or low flow condition with no underload protection (same as above)
- Rapid decreases in well bore pressure (can cause decompression damage of power cable, MLE, or penetrators)
- Increasing unit production quickly cause rapid inflow of sand or foreign material

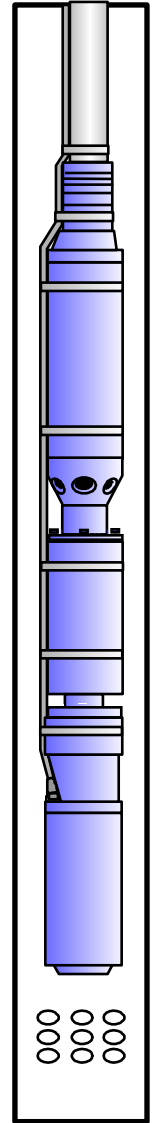


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Old Age

Old Age is the main reason for failure of ESP units world-wide. Typical reasons for pulling an old ESP unit are ...

- Low production due to pump wear
- Burned motor due to overload
- Burned motor due to fluid migration from seal section
- Down hole fault in the cable or motor lead due to decompression damage



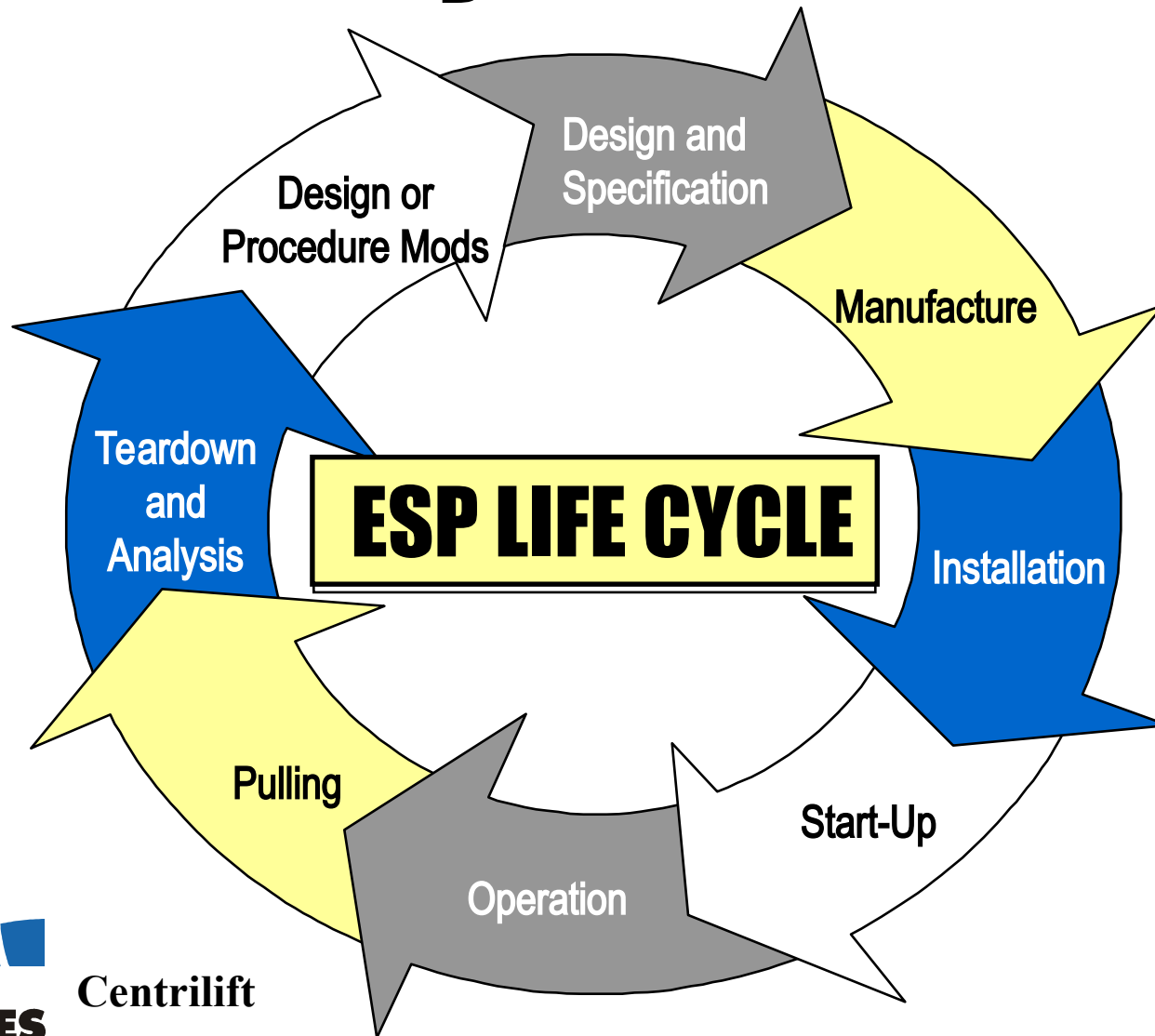
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ESP Operating Parameters and Troubleshooting



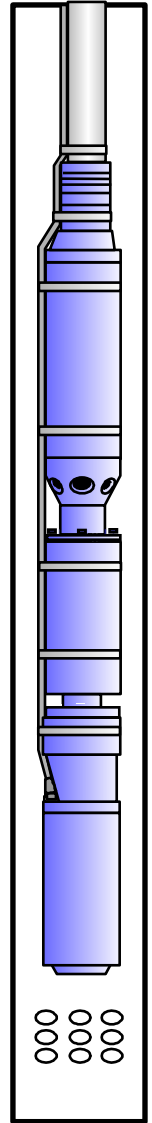
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The ESP Life Cycle



ESP Operation

- Once an ESP system is commissioned, the operator plays a key role in the system's performance and run life
- Key parameters must be monitored to insure proper operation of the system
- Failure to properly monitor or interpret these parameters can be costly

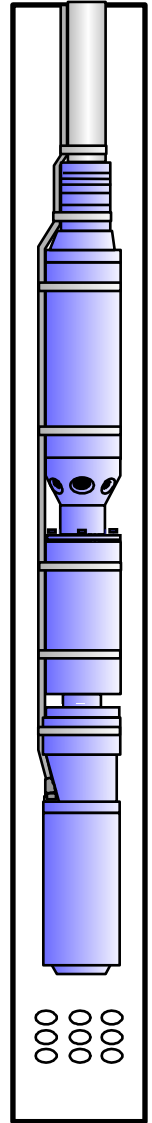


Operating Parameters

Three basic ESP operating parameters are ...

- Gross Production Rate
- Pump Intake Pressure
- Operating Motor Current

By monitoring these parameters, an Operator can better determine the relative condition of an ESP or anticipate possible problems



Production Rate

By monitoring the production rate an operator can ...

- Determine the approximate operating point on the pump curve
- Trend the rate of declining production
- Look for possible pump wear, tubing leaks, etc.

Loss of production is usually the first indicator of a downhole problem with an ESP



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Pump Intake Pressure

By monitoring PIP, an Operator can ...

- Determine relative unit sizing accuracy by comparing with the computer sizing
- Anticipate unit cycling
- Look for tubing leak, pump plugging and/or wear

Increases or decreases in PIP can indicate a change in the pump performance, well inflow, or installation integrity



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Motor Current

By monitoring motor current, an Operator can ...

- Look for trends in unit loading
- Spot possible motor damage due to electrical or mechanical problems
- Determine relative pump load or spot changes in loading
- Detect changes in downhole fluid condition

Changes in operating current indicate that the motor is reacting to a new input from the pump, well, or electrical system. The Motor Controller should shut the unit off if the current varies beyond acceptable limits

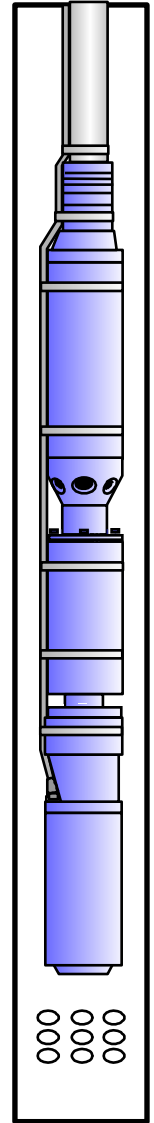


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Additional Operating Parameters

Other operating parameters that may be monitored include ...

- Pump Discharge Pressure
- Bottom Hole Temperature
- Discharge Fluid Temperature
- Motor Operating Temperature
- Unit Vibration



Troubleshooting

- Troubleshooting by an Operator involves looking at the unit operating parameters, as a group, to determine a possible cause
- By process of elimination, a cause and affect sequence can be developed when ESP operating problems occur
- Failure to check all parameters and/or call for assistance when required can result in premature failure of a unit

Troubleshooting Tools

Troubleshooting any system requires the proper tools. In the case of an ESP system this means information which includes ...

- Well history (including workovers, treatments, etc.)
- Previous ESP run life and failure modes
- Amp charts (prior to and during time of failure)
- Production data and historic trends
- Available bottom hole and surface pressure data
- Information on starts & stops or operator intervention

SPH, p.165-192



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